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Paper [406] — Continued

On the Curves which Satisfy Given Conditions.

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is situate on the nodal line of the torse, and that the quartic curve touches there the sheet which is not touched by the tangent plane $a = 0$; for this being so the quartic curve touching one sheet and simply meeting the other sheet meets the torse in three consecutive points, or the two points of intersection count each of them three times.

The torse has the cuspidal line

$$S = ae - 4bd + 3c^2 = 0, \quad T = ace + 2bcd - ad^2 - b^2e - c^3 = 0,$$

and the nodal line

$$\frac{6(ac - b^2), \quad 3(ad - bc), \quad ae + 2bd - 3c^2, \quad 3(be - cd), \quad 6(ce - d^2)}{a, \quad b, \quad c, \quad d, \quad e}$$

and the equations of the nodal line are satisfied by the values ($a = 0, b = 0, 3ce - 2d^2 = 0$) of the coordinates of the points in question.

To find the tangent planes at these points, starting from the equation $S^3 - 27T^2 = 0$ of the torse, taking (A, B, C, D, E) as current coordinates, and writing

$$\partial = A\partial_a + B\partial_b + C\partial_c + D\partial_d + E\partial_e,$$

then the equation of the tangent plane is in the first instance given in the form $S^2\partial S - 18T\partial T = 0$, which writing therein ($a = 0, b = 0, 3ce - 2d^2 = 0$) assumes, as it should do, the form $0 = 0$; the left-hand side is in fact found to be $9c^4(3ce - 2d^2)A$. Proceeding to the second derived equation, this is $S^2\partial^2 S + 2S(\partial S)^2 - 18T\partial^2 T - 18(\partial T)^2 = 0$, or substituting the values of the several terms, the equation is

$$\begin{aligned} &9c^4(AE - 4BD + 3C^2) + 3c^2(eA - 4dB + 6cC)^2 \\ &\quad + 18c^3\{e(AC - B^2) + 2d(BC - AD) + c(AE + 2BD - 3C^2)\} \\ &\quad - 9\{(ce - d^2)A + 2cdB - 3c^2C\}^2 = 0; \end{aligned}$$

the terms in BC, BD, C^2 vanish identically, that in B^2 is $(48 - 36) = 12c^3d^2 - 18c^4e, = -6c^4(3ce - 2d^2)B^2$, which also vanishes; hence there remain only the terms divisible by A , giving first the tangent plane

$A = 0$, and secondly the other tangent plane,

$$A(-6c^3d^2 + 18cd^2e - 9c^4) + B(-60c^3de + 36cd^3) + C(108c^3e - 54c^2d^2) + D(-36c^3d) + E \cdot 27c^4 = 0.$$

Taking the equations of the quadric surfaces to be

$$(\lambda, \mu, \nu, \rho, \sigma, \tau \chi a^2, b^2, ab, 3ce - 2d^2, ae - 8bd, ad - 12bc) = 0, \\ (\lambda', \mu', \nu', \rho', \sigma', \tau' \chi \dots\dots\dots) = 0,$$

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the equations of the tangent planes are

$$\rho(3cE + 5eC - 4dD) + \sigma(eA - 8dB) + \tau(dA - 12cB) = 0, \\ \rho'(\dots\dots\dots) + \sigma'(\dots\dots\dots) + \tau'(\dots\dots\dots) = 0,$$

in all which equations we have $3ce - 2d^2 = 0$; and if to satisfy this equation we write $c : d : e = 2 : 3\beta : 3\beta^2$, then the equations of the tangent planes become

$$\rho\beta^2(A\beta - 8B) + \sigma(3C\beta^2 - 4D\beta + 2E) = 0, \\ \rho(8B\beta - 4D\beta + 2E) + (\sigma\beta + \tau)(A\beta - 8B) = 0, \\ \rho'(\dots\dots\dots) + (\sigma'\beta - \tau')(\dots\dots\dots) = 0,$$

or the three tangent planes intersect in the line

$$A\beta - 8B = 0, \quad 3C\beta^2 - 4D\beta + 2E = 0,$$

which completes the proof.

Reverting to the sextic locus,

$$(ae + 4bd - 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

considered as a locus in 4-dimensional space depending on the five coordinates (a, b, c, d, e) , this has upon it the twofold locus

$$ae + 4bd + 3c^2 = 0, \quad ace + 2bcd - ad^2 - b^2e - c^3 = 0,$$

say the cuspidal locus, of the order 6, and the twofold locus

$$\frac{2(ac - b^2), 3(ad - bc), ae + 2bd - 3c^2, 3(be - cd), 2(ce - d^2)}{a, b, c, d, e} = 0,$$

say the nodal locus, of the order 4: there is also a threefold locus

$$\begin{vmatrix} a, & b, & c, & d \\ b, & c, & d, & e \end{vmatrix} = 0,$$

say the supercuspidal locus, of the order 4.

We thence at once infer

$$(1/l, 2) = 6, \text{ agreeing with } (1/l, 2) = \alpha - 4, \\ (1/l, 1, 1) = 4, \text{ agreeing with } (1/l, 1, 1) = 2m^2 + 2mn + \frac{1}{2}n^2 - 8m - 5n + 13 + \frac{1}{2}\alpha, \\ (1/l, 3) = 4, \text{ " } (1/l, 3) = -4m - 3n - 5 + 3\alpha;$$

but I have not investigated the application to the symbols with $\cdot/$ or $//$.

If the conic, instead of simply passing through the cusp, touches the cuspidal tangent, then in the equation $(a, b, c, f, g, h \chi x, y, z)^2 = 0$ of the conic we have $f = 0$, or, what is the same thing, in the equation $(e, bc, 0, \frac{1}{2}a, 2b, 2d \chi x, y, z)^2 = 0$ of the conic we have $a = 0$. The equation in θ is thus reduced to $4b\theta^3 + 6c\theta^2 + 4d\theta + e = 0$. For the independent discussion of this case it is convenient to alter the coefficients so that the equation in θ may be in the standard form $(a, b, c, d \chi \theta, 1)^3 = 0$, viz. we

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assume the equation of the conic to be $(d, 3b, 0, 0, \frac{1}{2}a, \frac{1}{2}c \chi x, y, z)^2 = 0$. The equation of the contact-locus then is

$$C^2c^2 + 4\Phi acd^2 + 4\Psi d^3 - 6abcd - 3\Psi^2c^2 = 0,$$

viz. this is a developable surface, or torse, of the order 4, and we at once infer $(2/l, 1) = 4$, agreeing with $(2/l, 1) = 2m + n - 5$.

I will show also that we have $(2/l, 1 \cdot /) = 6$, agreeing with $(2/l, 1 \cdot /) = 2m + 2n - 6$, and $(2/l, 1//) = 5$, $(2/l, 1//) = m + 2n - 4$.

The condition that the conic may touch an arbitrary line $\alpha x + \beta y + \gamma z = 0$, is in fact

$$(0, -\frac{1}{2}a, f(4bd - 3c^2), fac, -\frac{1}{2}ab, 0 \chi \alpha, \beta, \gamma)^2 = 0,$$

which, considering therein (a, b, c, d) as coordinates, is the equation of a quadric surface passing through the conic $a = 0, 4bd - 3c^2 = 0$; the quartic torse also passes through this conic; hence the quadric surface and the torse intersect in this conic, which is of the order 2, and in a residual curve of the order 6; and the number of the conics $(2/l, 1/)$ is equal to the order of this residual curve, that is, it is = 6.

If the conic touch a second arbitrary line $\alpha'x + \beta'y + \gamma'z = 0$, then we have in like manner the quadric surface

$$(0, -\frac{1}{2}a', f(4bd - 3c^2), fa'c, -\frac{1}{2}a'b, 0 \chi \alpha', \beta', \gamma')^2 = 0;$$

that is, we have the quartic torse and two quadric surfaces, each passing through the conic $a = 0, 4bd - 3c^2 = 0$, and it is to be shown that the number of intersections not on this conic is = 5.

The two quadric surfaces intersect in the conic and in a second conic; this second conic meets the torse in 8 points, but 2 of these coincide with the point $a = 0, b = 0, c = 0$, which is one of the intersections of the two conics (the point $a = 0, b = 0, c = 0$ is in fact a point on the cuspidal edge of the torse, and, the conic passing through it, reckons for 2 intersections), and 1 of the 8 points coincides with the other of the intersections of the two conics; there remain therefore $8 - 2 - 1 = 5$ intersections, or we have $(2/l, 1//) = 5$.

Annex No. 5 (referred to, Nos. 22 and 71).

On the Conics which have contact of the third order with a given cuspidal cubic, and two contacts (double contact) with a given conic.

Let the equation of the cuspidal cubic be $x^2z - y^3 = 0$ ($x =$ tangent at cusp, $z =$ tangent at inflexion, $y =$ line joining cusp and inflexion; equation satisfied by $x : y : z = 1 : \theta : \theta^3$); and let the equation of the given conic be $U = (a, b, c, f, g, h \chi x, y, z)^2 = 0$;

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viz. in the rational form this is

then writing

$$\Theta = (a, b, c, f, g, h \chi 1, \theta, \theta^3)^2 = c\theta^6 + 2f\theta^4 + 2g\theta^3 + b\theta^2 + 2h\theta + a,$$

the equation of a conic having with the given cubic at a given point $(1, \theta, \theta^3)$ contact of the second order, and having double contact with the given conic, is

$$\sqrt{\Theta} \cdot \begin{vmatrix} \sqrt{\Theta}, & x, & y, & z \\ \sqrt{\theta}, & 1, & \theta, & \theta^3 \\ (\sqrt{\theta})', & . & 1, & 3\theta^2 \\ (\sqrt{\theta})'', & . & . & 6\theta \end{vmatrix} = 0,$$

viz. in the rational form this is

$$\Theta = \begin{vmatrix} \sqrt{\Theta}, & x, & y, & z \\ \sqrt{\theta}, & 1, & \theta, & \theta^3 \\ (\sqrt{\theta})', & . & 1, & 3\theta^2 \\ (\sqrt{\theta})'', & . & . & 6\theta \end{vmatrix}^2 = 0,$$

and this will have at the point $(1, \theta, \theta^3)$ a contact of the third order if θ be determined by

$$\begin{vmatrix} \sqrt{\Theta}, & 1, & \theta, & \theta^3 \\ (\sqrt{\theta})', & . & 1, & 3\theta^2 \\ (\sqrt{\theta})'', & . & . & 6\theta \\ (\sqrt{\theta})''', & . & . & 6 \end{vmatrix} = 0,$$

viz. this is $\Theta \cdot (\sqrt{\theta})''' - (\sqrt{\Theta})''' = 0$; or developing and multiplying by θ^6 this is

$$\Theta(2\theta\theta''' - \theta'') + \theta\theta'(-\frac{3}{2}\theta\theta''' + \frac{1}{2}\theta'') + \theta'^2 \cdot \frac{1}{2}\theta\theta' = 0;$$

and substituting for Θ its value, this is

$$\begin{aligned} & (c\theta^6 + 2f\theta^4 + 2g\theta^3 + b\theta^2 + 2h\theta + a)^2(45c\theta^4 + 12f\theta^2 - 6) \\ & \quad + (c\theta^6 + 2f\theta^4 + 2g\theta^3 + b\theta^2 + 2h\theta + a) \\ & (3c\theta^5 + 4f\theta^3 + 3g\theta^2 + b\theta + h)(-42c\theta^4 - 32f\theta^2 - 16g\theta - 2h) \\ & \quad + \theta^6(3c\theta^5 + 4f\theta^3 + 3g\theta^2 + b\theta + h)^2 = 0. \end{aligned}$$

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The coefficients of the powers 16, 15, 14, 13 of θ all vanish, so that this is in fact an equation of the twelfth order $(2/l, 1//) = 0$; and putting, as usual,

$$(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch) = (A, B, C, F, G, H),$$

the equation is found to be

$$\begin{aligned} & -4cA \cdot \theta^{12} \\ & + 7hA \cdot \theta^{11} + (\frac{1}{2}aB + 8cH)\theta^{10} + (9gB - 20fC)\theta^9 \\ & - 36cB \cdot \theta^8 - 22bH \cdot \theta^7 + (16f^2 + 40aA)\theta^6 \\ & + 10gG \cdot \theta^5 - 10cG \cdot \theta^4 - 180hH \cdot \theta^3 + 20aG \cdot \theta^2 \\ & - 12aF \cdot \theta + 40g^2 + 20b^2 + 40fF - 60fB \cdot \theta^5 \\ & + 8hB \cdot \theta^4 - abc \cdot \theta^3 - 90g^2 + 2bG \cdot \theta^2 - aC = 0, \end{aligned}$$

where the form of the coefficients may be modified by means of the identical equations

$$\begin{aligned} & (A, H, G \chi a, h, g)^2 = K, \quad (H, B, F \chi a, h, g)^2 = 0, \quad (G, F, C \chi a, h, g)^2 = 0, \\ & (A, H, G \chi h, b, f)^2 = 0, \quad (H, B, F \chi h, b, f)^2 = K, \quad (G, F, C \chi h, b, f)^2 = 0, \\ & (A, H, G \chi g, f, c)^2 = 0, \quad (H, B, F \chi g, f, c)^2 = 0, \quad (G, F, C \chi g, f, c)^2 = K. \end{aligned}$$

There is consequently a conic answering to each value of θ .

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Substituting these values, but retaining (a, b, c, f, g, h) as standing for their values $a = 2\lambda\lambda'$, &c., the equation in θ is found to contain the cubic factor $2X\theta^3 - 3Y\theta^2 + Z$, where it is to be observed that this factor equated to zero determines the values of θ which correspond to the points of contact with the cuspidal cubic of the tangents from the point (X, Y, Z) , which is the intersection of the lines $\lambda x + \mu y + \nu z = 0$, and $\lambda'x + \mu'y + \nu'z = 0$; and omitting the cubic factor, the residual equation is found to be $(2/l, 1//) = 0$, where the form of the coefficients may be modified by means of the identical equations

$$aX + hY + gZ = 0, \quad hX + bY + fZ = 0, \quad gX + fY + cZ = 0.$$

The equation is of the 9th order, and there are consequently 9 conics.

Annex No. 6 (referred to, No. 48).

Containing, with the variation referred to in the text, Zeuthen's forms for the characteristics of the conics which satisfy four conditions.

(1)

$$\begin{aligned}
 (1) &= \eta + 2m, \\
 (1, 1) &= \eta + 2m, \quad (1 \cdot /) = 2m - 4m, \quad (1//) = 4\eta + 4m, \quad (1///) = 4\eta + 2m, \quad (1111) = 2\eta + m; \\
 (2) &= 2m(m + n - 3) + \tau, \\
 (2, 1) &= 2m(m + 2n - 5) + 2\tau, \quad (2//) = 2\eta(2m + n - 5) + 2\delta, \\
 (2//) &= 2\eta(m + n - 3) + \delta; \\
 (3) &= \frac{1}{2}[2m^2 + 6m^2n - n^2 - 30m^2 - 18mn + 13n^2 + 84m - 42n + (6m + 3n - 26)\tau], \\
 (3 \cdot /) &= \frac{1}{2}[(m + n)(-(m + n)^2 - 7(m + n) + 48) + 4mn(3m + 3n - 13) + 2(3m + 3n - 20)(\delta + \tau)], \\
 (3//) &= \frac{1}{2}[-m^2 + 6mn^2 + 2n^2 + 18m^2 - 18mn - 30n^2 - 42m + 84n + (3m + 6n - 26)\delta].
 \end{aligned}$$

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$$\begin{aligned}
 (1, 1, 1, 1) &= \frac{1}{2}\{2(m - 3)(m - 4)(\eta^2 - m - n) + (n - 3)(n - 4)(m^2 - m - \eta) \\
 &\quad + 4(m^2 - 11m + 28)\tau + 2(n^2 - 11n + 28)\delta \\
 &\quad + (4(m - 4)(n - 4) - 1)(2\delta + \tau) + 2\delta^2 + \tau^2\}, \\
 (1 \cdot /) &= \frac{1}{2}\{(m - 3)(m - 4)(\eta^2 - m - n) + 2(n - 3)(n - 4)(m^2 - m - \eta) \\
 &\quad + 2(m^2 - 11m + 28)\tau + 4(n^2 - 11n + 28)\delta \\
 &\quad + (4(n - 4)(m - 4) - 1)(\delta + 2\tau) + \delta^2 + 2\tau^2\};
 \end{aligned}$$

(2, 1)

$$\begin{aligned}
 (2, 1) &= 3m + \iota, \\
 (2 \cdot 1) &= 2(3m + \iota), \quad (2, 1//) = 2(3m + \iota), \\
 (2, 1//) &= 3(2mn + n^2 + 4m - 10n) + (2m + n - 14)\kappa, \\
 (2, 1, 1) &= 2(3m + \iota)(m + n - 12) + 24(m + n), \\
 (2 \cdot 1, 1) &= 3(m^2 + 2mn - 10m + 4\eta) + (m + 2n - 14)\tau; \\
 (2 \cdot /) &= (2m + n - 7)(6\delta + (n - 3)\kappa) + \{(m - n)(m + n - 5) + \tau\}(3m + \iota - 36) + 12(m - n)(m + n - 3), \\
 (2//) &= (m + 2n - 7)(6\delta + (m - 3)\iota) + \{(n - m)(m + n - 5) + \delta\}(3m + \iota - 36) + 12(n - m)(m + n - 3);
 \end{aligned}$$

(2, 2)

$$\begin{aligned}
 (2, 2) &= 6n - 4m + 3\iota = 5m - 3n + 3\iota, \\
 (2 \cdot 2) &= 10n - 8m + 6\iota = 10m - 8n + 6\iota, \\
 (2//) &= 5n - 3m + 3\kappa = 6m - 4n + 3\tau; \\
 (3) &= 2(-4m^2 + 3mn + 3n^2 + 28m - 32n) + 3(2m + n - 13)\kappa, \\
 (3 \cdot /) &= 2(3m^2 + 3mn - 4n^2 - 32m + 28n) + 3(m + 2n - 13)\tau; \\
 (4) &= 10n - 10m + 6\iota = 8m - 8n + 6\iota, \\
 (4//) &= 10n - 8m + 6\kappa = 10m - 10n + 6\iota.
 \end{aligned}$$

Page 258**Annex No. 7** (referred to, No. 93).

In connexion with De Jonquières' formula.

I have been led to consider the following question. Given a set of equations:

$$\begin{aligned}
 a &= a \quad (\text{viz. } b = b, c = c, \&c.), \\
 ab &= ab \quad (\text{viz. } ac = ac, \&c., \text{ and the like in all the subsequent equations}), \\
 &+ (11) a \cdot b; \\
 abc &= abc + (12) (a \cdot bc + b \cdot ac + c \cdot ab) + (111) a \cdot b \cdot c, \\
 abcd &= abcd + (13) (a \cdot bcd + \&c.) + (22) (ab \cdot cd + \&c.) \\
 &+ (112) (a \cdot b \cdot cd + \&c.) + (1111) a \cdot b \cdot c \cdot d,
 \end{aligned}$$

and so on indefinitely (where the () is used to denote multiplication, and $ab, abc, \&c.$, and also $a \cdot b, a \cdot b \cdot c, \&c.$ are so many separate and distinct symbols not expressible in terms of $a, b, c, \&c.$, $a, b, c, \&c.$), then we have conversely a set of equations

$$\begin{aligned}
 a &= a \quad (\text{viz. } b = b, c = c, \&c.), \\
 ab &= ab \quad (\text{viz. } ac = ac, \&c., \text{ and the like in all the subsequent equations}) \\
 &+ [11] a \cdot b; \\
 abc &= abc + [12] (a \cdot bc + b \cdot ac + c \cdot ab) + [111] a \cdot b \cdot c, \\
 abcd &= abcd + [13] (a \cdot bcd + \&c.) + [22] (ab \cdot cd + \&c.) \\
 &+ [112] (a \cdot b \cdot cd + \&c.) + [1111] a \cdot b \cdot c \cdot d,
 \end{aligned}$$

and so on; and it is required to find the relation between the coefficients () and []; we find, for example,

$$\begin{aligned}
 [11] &= -(11), \\
 [12] &= -(12), \\
 [111] &= 3(11)(12) - (111), \\
 [13] &= -(13), \\
 [22] &= -(22), \\
 [112] &= 2(13)(12) + (22)(11) - (112), \\
 [1111] &= -12(13)(12)(11) + 4(13)(111) + 3(22)(11)(11) - 6(112)(11) - (1111);
 \end{aligned}$$

and it is to be noticed that, conversely, the coefficients () are given in terms of the coefficients [] by the like equations with the very same numerical coefficients; in fact from the last set of equations, this is at once seen to be the case as far as (112); and for the next term (1111) we have

$$\begin{aligned}
 (1111) &= +12[13][12][11] - 4[13](3[12][11] - [111]) + 3[22][11][11] - 6[11](2[13][12]) \\
 &+ [22][11][11] - [1111] \\
 &= +12[13][12][11] - 4[13][111] - 3[22][11][11] + 6[112][11] - [1111],
 \end{aligned}$$

having the same coefficients $-12, +4, -3, +6, -1$ as in the formula for [1111] in terms of the coefficients (); it is easy to infer that the property holds good generally.

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To explain the law for the expression of the coefficients of either set in terms of the other set, I consider, for example, the case where the sum of the numbers in the (), or [] is $= 5$; and I form a kind of tree as follows:

[Tree diagram showing symbolic partitions]

the formation of which is obvious; and I derive from it in the manner about to be explained the expressions for the coefficients [14], [23] &c. in terms of the corresponding coefficients in (); viz. we

have

$$\begin{aligned}
[14] &= -(14), \\
[23] &= -(23), \\
[113] &= 2(14)(13) + (23)(11) - (113), \\
[122] &= (14)(22) + 2(23)(12) - (112), \\
[1112] &= -6(14)(13)(12) - 3(14)(22)(11) + 3(14)(112) - 6(23)(12)(11) \\
&\quad + 3(113)(12) + (23)(111) + 3(122)(11) - (1112), \\
[11111] &= +60(14)(13)(12)(11) - 20(14)(13)(111) + 15(14)(22)(11)(11) \\
&\quad - 30(14)(112)(11) + 5(14)(1111) + 30(23)(12)(11)(11) \\
&\quad - 10(23)(111)(11) - 30(113)(12)(11) + 10(113)(111) \\
&\quad - 15(122)(11)(11) + 10(1112)(11) - (11111).
\end{aligned}$$

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To form the symbolic parts, we follow each branch of the tree to each point of its course: thus from the branch 113 we have (113) belonging to [113], (113)(111) belonging to [11111], (113)(12) belonging to [1112], (113)(12)(11) belonging to [11111]; viz. (113) belongs to [113]; (113)(111), read 11 (3 replaced by) 111, belongs to [11111]; (113)(12), read 11 (3 replaced by) 12, belongs to [1112]; (113)(12)(11), read 11 (3 replaced by) 1 (2 replaced by) 11, belongs to [11111].

And observe that where (as, for example, with the symbol 122) there are branches derived from two or more figures, we pursue each such branch separately, and also all or any of them simultaneously to every point in the course of such branch or branches; thus for the branch 122 we have (122) belonging to [122], (122)(11) belonging to [1112], (122)(11)(11) belonging to [11111].

Similarly for the branch 23 we have (23) belonging to [23], (23)(111) belonging to [1112], (23)(12) belonging to [122], (23)(12)(11) (same as infra) belonging to [1112], (23)(11)(111) belonging to [11111], (23)(11)(12) (same as supra) belonging to [1112], (23)(11)(12)(11) belonging to [11111].

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We thus obtain the symbolic parts of the several expressions for [14], [23], [11111] respectively: the sign of each term is + or – according as the number of factors in () is even or odd; thus in the expression for [11111], the term (14)(13)(12)(11) has four factors, and is therefore +, the term (113)(12)(11) has three factors, and is therefore –.

The numerical coefficients are obtained as follows. There is a common factor derived from the expression in []

Paper [407]

Second Memoir on the Curves which Satisfy Given Conditions; The Principle of Correspondence.

[PAGES 263–295]

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLVIII. (for the year 1868), pp. 145–172. Received April 18,—Read May 2, 1867.]

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In the present Memoir I reproduce with additional developments the theory established in my paper “On the Correspondence of two points on a Curve” (London Math. Society, No. VII., April 1866), [385]; and I endeavour to apply it to the determination of the number of the conics which satisfy given conditions; viz. these are conditions of contact with a given curve, or they may include arbitrary conditions Z , $2Z$, &c.

If, for a moment, we consider the more general question where the Principle is to be applied to finding the number of the curves C^r of the order r , which satisfy given conditions of contact with a given curve, there are here two kinds of special solutions; viz., we may have proper curves C^r touching (specially) the given curve at a cusp or cusps thereof, and we may have improper curves, that is, curves which break up into two or more curves of inferior orders. In the case where the curves C^r are lines, there is only the first kind of special solution, where the sought for lines touch at a cusp or cusps. But in the case to which the Memoir chiefly relates, where the curves C^r are conics, we have the two kinds of special solutions, viz., proper conics touching at a cusp or cusps, and conics which are line-pairs or point-pairs. In the application of the Principle to determining the number of the conics which satisfy any given conditions, I introduce into the equation a term called the “Supplement” (denoted by the abbreviation “Supp.”), to include the special solutions of both kinds. The expression of the Supplement should in every case be furnished by the theory; and this being known, we should then have an equation leading to the number of the conics which properly satisfy the prescribed conditions; but in thus finding the expression of the Supplements, there are difficulties which I am unable to overcome; and I have contented myself with the reverse course, viz., knowing in each case the

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number of the proper solutions, I use these results to determine *a posteriori* in each case the expression of the Supplement; the expression so obtained can in some cases be accounted for readily enough, and the knowledge of the whole series of them will be a convenient basis for ulterior investigations.

The Principle of Correspondence for points in a line was established by Chasles in the paper in the *Comptes Rendus*, June–July 1864, referred to in my First Memoir; it is extended to unicursal curves in a paper of the same series, March 1866, “*Sur les courbes planes ou à double courbure dont les points peuvent se déterminer individuellement—Application du Principe de Correspondance dans la théorie de ces courbes*,” but not to the case of a curve of given deficiency D considered in my paper of April 1866 above referred to.

The fundamental theorem in regard to unicursal curves, viz. that in a curve of the order m with $\frac{1}{2}(m-1)(m-2)$ double points (nodes or cusps) the coordinates (x, y, z) are proportional to rational and integral functions of a variable parameter θ ,—as a case of a much more general theorem of Riemann’s—dates from the year 1857, but was first explicitly stated by Clebsch in the paper “*Ueber diejenigen ebenen Curven deren Coordinaten rationale Functionen eines Parameters sind*,” *Crelle*, t. LXIV. (1864), pp. 43–63. See also my paper “On the Transformation of Plane Curves,” *London Mathematical Society*, No. III., Oct. 1865, [384].

The paragraphs of the present Memoir are numbered consecutively with those of the First Memoir.

Article Nos. 94 to 104. On the Correspondence of two points on a Curve.

94. In a unicursal curve the coordinates (x, y, z) of any point thereof are proportional to rational and integral functions of a variable parameter θ . Hence if two points of the curve correspond in such wise that to a given position of the first point there correspond α' positions of the second point, and to a given position of the second point α positions of the first point, the number of points which correspond each to itself is $= \alpha + \alpha'$.

For let the two points be determined by their parameters θ, θ' respectively, then to a given value of θ there correspond α' values of θ' , and to a given value of θ' there correspond α values of θ ; hence the relation between (θ, θ') is of the form $(\theta, 1)^\alpha (\theta', 1)^{\alpha'} = 0$; and writing therein $\theta' = \theta$, then for the points which correspond each to itself, we have an equation $(\theta, 1)^{\alpha+\alpha'} = 0$, of the order $\alpha + \alpha'$; that is, the number of these points is $= \alpha + \alpha'$.

Hence for a unicursal curve we have a theorem similar to that of M. Chasles' for a line, viz. the theorem may be thus stated:

If two points of a unicursal curve have an (α, α') correspondence, the number of united points is $= \alpha + \alpha'$.

But a unicursal curve is nothing else than a curve with a deficiency $D = 0$, and we thence infer:

Theorem. *If two points of a curve with deficiency D have an (α, α') correspondence, the number of united points is $= \alpha + \alpha' + 2kD$; in which theorem $2k$ is a coefficient to be determined.*

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95. Suppose that the corresponding points are P, P' and imagine that when P is given the corresponding points P' are the intersections of the given curve by a curve Θ (the equation of the curve Θ will of course contain the coordinates of P as parameters, for otherwise the position of P' would not depend upon that of P). I find that if the curve has with the given curve k intersections at the point P , then in the system of points (P, P') the number of united points is $\alpha = \alpha + \alpha' + 2kD$, whence in particular if the curve Θ does not pass through the point P , then the number of united points is $= \alpha + \alpha'$, as in the case of a unicursal curve.

(I have in the paper of April 1866 above referred to, proved this theorem in the particular case where the k intersections at the point P take place in consequence of the curve Θ having a k -tuple point at P , but have not gone into the more difficult investigation for the case where the k intersections arise wholly or in part from a contact of the curve Θ , or any branch or branches thereof, with the given curve at P .)

96. It is to be observed that the general notion of a united point is as follows: taking the point P at random on the given curve, the curve Θ has at this point k intersections with the given curve; the remaining intersections are the corresponding points P' ; if for a given position of P one or more of the points P' come to coincide with P , that is, if for the given position of P the curve Θ has at this point more than k intersections with the given curve, then the point in question is a united point.

It might at first sight appear that if for a given position of P a number 2, 3, ... or j of the points P' should come to coincide with P , then that the point in question should reckon, for 2, 3, ... or j (as the case may be) united points: but this is not so. This is perhaps most easily seen in the case of a unicursal curve; taking the equation of correspondence to be $(\theta, 1)^\alpha (\theta', 1)^{\alpha'} = 0$, then we have $\alpha + \alpha'$ united points corresponding to the values of θ which satisfy the equation $(\theta, 1)^\alpha (\theta, 1)^{\alpha'} = 0$; if this equation has a j -tuple root $\theta = \lambda$, the point P which answers to this value λ of the parameter is reckoned as j united points.

But starting from the equation $(\theta, 1)^\alpha (\theta', 1)^{\alpha'} = 0$, if on writing in this equation $\theta = \lambda$, the resulting equation $(\lambda, 1)^\alpha (\theta', 1)^{\alpha'} = 0$ has a root $\theta' = \lambda$, it follows that the equation $(\theta, 1)^\alpha (\theta, 1)^{\alpha'} = 0$ has a root $= \lambda$, and that the point which belongs to the value $= \lambda$ is a united point; if on writing in the equation $\theta' = \lambda$, the resulting equation $(\lambda, 1)^\alpha (\theta', 1)^{\alpha'} = 0$ has a j -tuple root $\theta' = \lambda$, it does not follow that the equation $(\theta, 1)^\alpha (\theta, 1)^{\alpha'} = 0$ has a j -tuple root $= \lambda$, nor consequently that the point answering to $= \lambda$ in anywise reckons as j united points.

97. This may be further illustrated by regarding the parameters θ, θ' as the coordinates of a point in a plane; the equation $(\theta, 1)^\alpha (\theta', 1)^{\alpha'} = 0$ is that of a curve of the order $\alpha + \alpha'$, having an α -tuple point at infinity on the axis $\theta = 0$, and an α' -tuple point at infinity on the axis $\theta' = 0$; the united points are given as the intersections of the curve with the line $\theta = \theta'$; a j -fold intersection, whether arising from

a multiple point of the curve or from a contact of the line $\theta = \theta'$ with the curve, gives a point which reckons as j united points.

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But if $\theta = \lambda$ gives the j -fold root $\theta' = \lambda$, this shows that the line $\theta = \lambda$ has with the curve j intersections at the point $\theta = \theta' = \lambda$; not that the line $\theta = \theta'$ has with the curve j intersections at the point in question.

98. Reverting to the notion of a united point as a point P which is such that one or more of the corresponding points P' come to coincide with P ; in the case where P is at a node of the given curve, it is necessary to explain that the point P must be considered as belonging to one or the other of the two branches through the node, and that the point P is not to be considered as a united point unless we have on the same branch of the curve one or more of the corresponding points P' coming to coincide with the point P .

If, to fix the ideas, $k = 1$, that is, if the curve Θ simply pass through the point P , then if P be at a node the curve passes through the node and has therefore at this point two intersections with the given curve; but the second intersection belongs to the other branch, and the node is not a united point; in order to make it so, it is necessary that the curve should at the node touch the branch to which the point P is considered to belong.

The thing appears very clearly in the case of a unicursal curve; we have here two values $\theta = \lambda$, $\theta = \lambda'$ answering to the node according as it is considered as belonging to one or the other branch of the curve; and in the equation of correspondence $(\theta, 1)^\alpha(\theta', 1)^{\alpha'} = 0$, writing $\theta = \lambda$, we have an equation $(\lambda, 1)^\alpha(\theta', 1)^{\alpha'} = 0$ satisfied by $\theta' = \lambda'$ but not by $\theta' = \lambda$, and the equation $(\theta, 1)^\alpha(\theta, 1)^{\alpha'} = 0$ is thus not satisfied by the value $\theta = \lambda$. The conclusion is that a node *quâ* node is not a united point.

99. But it is otherwise as regards a cusp. When the point P is at a cusp, the curve Θ (which has in general with the given curve k intersections at P) has

here more than k intersections, and (as in this case there is no distinction of branch) the cusp reckons as a united point. In the case of a unicursal curve, there is at the cusp a single value $\theta = \lambda$ of the parameter, and the equation $(\theta, 1)^\alpha(\theta, 1)^{\alpha'} = 0$ is satisfied by the value $\theta = \lambda$.

But for the very reason that the cusp *quâ* cusp reckons as a united point, the cusp is a united point only in an improper or special sense, and it is to be rejected from the number of true united points. We may include the cusps, along with any other special solutions which may present themselves, under a head “Supplement,” and instead of writing as above $\alpha - \alpha - \alpha' = 2kD$, write

$$\alpha - \alpha - \alpha' + \text{Supp.} = 2kD.$$

Before going further I apply the theorem to some examples in which the curve is a system of lines.

100. *Investigation of the class of a curve of the order m with δ nodes and κ cusps.*

Take as corresponding points on the given curve two points such that the line joining them passes through a fixed point; the united points will be the points of contact of the tangents through; that is, the number of the united points will be equal to the class of the curve.

The curve Θ is here the line OP which has with the given curve a single intersection at P ; that is, we have $k = 1$. The points P' corresponding to a given position of P are the remaining $m - 1$ intersections of OP with the curve, that is, we have $\alpha' = m - 1$; and in like manner $\alpha = m - 1$.

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Each of the cusps is (specially) a united point, and counts once, whence the Supplement is $= \kappa$. Hence, writing n for the class, we have

$$n + 2(m - 1) + \kappa = 2kD$$

or writing for $2D$ its value $= m^2 - 3m + 2 - 2\delta - 2\kappa$, we have $n = m^2 - m - 2\delta - 3\kappa$, which is right.

101. *Investigation of the number of inflexions.*

Taking the point P' to be a tangential of P (that is, an intersection of the curve by the tangent at P), the united points are the inflexions; and the number of the united points is equal to the number of the inflexions.

The curve Θ is the tangent at P having with the given curve two intersections at this point; that is, $k = 2$; P' is any one of the $m - 2$ tangentials of P , that is, $\alpha' = m - 2$; and P is the point of contact of any one of the $n - 2$ tangents from P' to the curve, that is, $\alpha = n - 2$.

Each cusp is (specially) a united point, and counts once, whence the Supplement is $= \kappa$. Hence, writing ι for the number of inflexions, we have

$$\iota - (m - 2) - (n - 2) + \kappa = 4D;$$

or substituting for $2D$ its value expressed in the form $n - 2m + 2 + \kappa$, we have

$$\iota = 3n - 2m + \kappa,$$

which is right.

102. For the purpose of the next example it is necessary to present the fundamental equation under a more general form. The curve Θ may intersect the given curve in a system of points P' , each p times, a system of points Q' , each q times, &c. in such manner that the points (P, P') , the points (P, Q') , &c. are pairs of points corresponding to each other according to distinct laws; and we shall then have the numbers $(\alpha, \alpha, \alpha')$, (b, β, β') , &c., corresponding to these pairs respectively, viz. (P, P') are points having an (α, α') correspondence, and the number of united points is $= a$; (P, Q') are points having a (β, β') correspondence, and the number of united points is $= b$, and so on. The theorem then is

$$p(a - \alpha - \alpha') + q(b - \beta - \beta') + \&c. + \text{Supp.} = 2kD,$$

being in fact the most general form of the theorem for the correspondence of two points on a curve, and that which will be used in all the investigations which follow.

103. Investigation of the number of double tangents.

Take P' an intersection of the curve with a tangent from P to the curve (or, what is the same thing, P, P' cotangentials of any point of the curve): the united points are here the points of contact of the several double tangents of the curve; or if τ be the number of double tangents, then the number of united points is $= 2\tau$.

The curve Θ is the system of the $n - 2$ tangents from P to the curve; each tangent has with the curve a single intersection at P , that is, $k = n - 2$; each tangent besides meets the curve in the point of contact Q' twice, and in $(m - 3)$ points P' ; hence if (a, α, α') refer to the points (P, Q') , and $(2\tau, \beta, \beta')$ to the points (P, P') , we have

$$2(a - \alpha - \alpha') + (2\tau - \beta - \beta') + \text{Supp.} = 2(n - 2)D.$$

From the foregoing example the value of $a - \alpha - \alpha'$ is $= 4D - \kappa$.

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In the case where the point P is at a cusp, then the $n - 2$ tangents become the $n - 3$ tangents from the cusp, and the tangent at the cusp; hence the curve Θ meets the given curve in $2(n - 3) + 3 = 2n - 3$ points, that is, $(n - 2) + (n - 1)$ points; this does not prove (ante, No. 96), but the fact is, that the cusp counts in the Supplement $(n - 1)$ times, and the expression of the Supplement is $= (n - 1)\kappa$.

It is clear that we have $\beta = \beta' = (n - 2)(m - 3)$, so that the equation is

$$8D - 2\kappa + 2\tau - 2(n - 2)(m - 3) + (n - 1)\kappa = (n - 2)2D,$$

that is

$$2\tau = 2(n - 2)(m - 3) + (n - 6)2D + (-n + 3)\kappa;$$

or substituting for $2D$ its value $= n - 2m + 2 + \kappa$ and reducing, this is

$$2\tau = n^2 + 4m - 10n - 3\kappa,$$

which is right.

104. As another example, suppose that the point P on a given curve of the order m and the point Q on a given curve of the order m' have an (α, α') correspondence, and let it be required to find the class of the curve enveloped by the line PQ .

Take an arbitrary point O , join OQ , and let this meet the curve m in P' ; then (P, P') are points on the curve m having a $(m'\alpha, m\alpha')$ correspondence; in fact to a given position of P there correspond α' positions of Q , and to each of these m' positions of P' ; that is, to each position of P there correspond $m\alpha'$ positions of P' ; and similarly to each position of P' there correspond $m'\alpha$ positions of P .

The curve Θ is the system of the lines drawn from each of the α' positions of Q to the point O , hence the curve Θ does not pass through P , and we have $k = 0$. Therefore the number of the united points (P, P') , that is, the number of the lines PQ which pass through the point O , is $= m\alpha' + m'\alpha$, or this is the class of the curve enveloped by PQ .

It is to be noticed that if the two curves are curves in space (plane, or of double curvature), then the like reasoning shows that the number of the lines PQ which meet a given line is $= m\alpha' + m'\alpha$, that is, the order of the scroll generated by the line PQ is $= m\alpha' + m'\alpha$.

Article Nos. 105 to 111. Application to the Conies which satisfy given conditions, one at least arbitrary.

105. Passing next to the equations which relate to a conic, we seek for $(4Z)(1)$, the number of the conics which satisfy any four conditions $4Z$ and besides touch a given curve, $(3Z)(2)$ and $(3Z)(1, 1)$, the number of the conics which satisfy three conditions, and besides have with the given curve a contact of the second order, or (as the case may be) two contacts of the first order; and so on with the conditions $2Z$, Z , and then finally (5) , $(4, 1)$, $\dots$ $(1, 1, 1, 1, 1)$, the numbers of the conics which have with the given curve a contact of the fifth order, or a contact of the fourth and also of the first order. $\dots$, or five contacts of the first order.

106. As regards the case $(4Z)(1)$, taking P an arbitrary point of the given curve m , and for the curve Θ the system of the conics $(4Z)(1)$ which pass through the given point P and besides satisfy the four conditions, then the curve has with the given curve $(4Z)(1)$ intersections at P , and the points P' are the remaining $(2m - 1)(4Z)(1)$ intersections: in the case of a united point (P, P') , some one of the system of conics becomes a conic $(4Z)(1)$; and the number of the united points is consequently equal to that of the conics $(4Z)(1)$; we have thus the equation

$$\{(4Z)(1) - 2(2m - 1)(4Z)(1)\} + \text{Supp. } (4Z)(1) = (4Z)(1) \cdot 2D.$$

107. It is in the present case easy to find *a priori* the expression for the Supplement.

1°. The system of conics $(4Z)$ contains $2(4Z) - (4Z/)$ point-pairs¹; each of these, regarded as a line, meets the given curve in m points, and each of these points is (specially) a united point (P, P') ; this gives in the Supplement the term $m\{2(4Z) - (4Z/)\}$.

2°. The number of the conics $(4Z)$ which can be drawn through a cusp of the given curve is $= (4Z)$; and the cusp is in respect of each of these conics a united point; we have thus the term $\kappa(4Z)$, and the Supplement is thus $= m\{2(4Z) - (4Z/)\} + \kappa(4Z)$.

We have moreover $(4Z)(1) = (4Z\cdot)$, $2D = n - 2m + 2 + \kappa$; and substituting these values, we find

$$\begin{aligned} (4Z)(1) &= (4m - 2)(4Z) \\ &\quad - m\{2(4Z) - (4Z/)\} - \kappa(4Z) + (n - 2m + 2 + \kappa)(4Z) \\ &= m(4Z/) + m(4Z/), \end{aligned}$$

which is right.

108. It is clear that if, instead of finding as above the expression of the Supplement, the value of $(4Z)(1)$, $= m(4Z/) + m(4Z/)$, had been taken as known, then the equation would have led to $\text{Supp. } (4Z)(1) = m\{2(4Z) - (4Z/)\} + \kappa(4Z)$; and this, as in fact already remarked, is the course of treatment employed in the remaining cases.

¹The expression a point-pair is regarded as equivalent to and standing for that of a coincident line-pair: see First Memoir, No. 30.

It is to be observed also that the equation may for shortness be written in the form

$$(4Z)\{(1) - 2(2m - 1)(1)\} + \text{Supp. } (1) = (1) \cdot 2D;$$

viz. the $(4Z)$ is to be understood as accompanying and forming part of each symbol; and the like in other cases.

109. We have the series of equations

$$\begin{aligned} (2Z)\{(2) - (2)(2m - 2) - (1, 1)\} + \text{Supp. } (2) &= (2) \cdot 2D; \\ (2Z)\{2(2) - (1, 1) - (2)(2m - 2)\} \\ &+ \{2(1, 1) - (1, 1)(2m - 3) - (1, 1)(2m - 3)\} + \text{Supp. } (1, 1) = (1, 1) \cdot 2D; \\ (2Z)\{(3) - (3)(2m - 3) - (2, 1)\} + \text{Supp. } (3) &= (3) \cdot 2D; \\ (2Z)\{2(3) - (2, 1) - (2, 1)\} \\ &+ \{(2, 1) - (2, 1)(2m - 4) - (1, 1, 1) \cdot 2\} + \text{Supp. } (2, 1) = (2, 1) \cdot 2D; \\ (2Z)\{3(3) - (1, 2) - (3)(2m - 3)\} \\ &+ \{(1, 2) - (1, 2)(2m - 4) - (1, 2)(2m - 4)\} + \text{Supp. } (1, 2) = (1, 2) \cdot 2D; \\ (2Z)\{2(2, 1) - (1, 1, 1) \cdot 2 - (2, 1)(2m - 4)\} \\ &+ \{3(1, 1, 1) - (1, 1, 1)(2m - 5) - (1, 1, 1)(2m - 5)\} \\ &+ \text{Supp. } (1, 1, 1) = (1, 1, 1) \cdot 2D; \end{aligned}$$

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$$\begin{aligned} (Z)\{(4) - (4)(2m - 4) - (1, 3)\} + \text{Supp. } (4) &= (4) \cdot 2D; \\ (Z)\{2(4) - (3, 1) - (2, 2)\} \\ &+ \{(3, 1) - (3, 1)(2m - 5) - (1, 1, 2)\} + \text{Supp. } (3, 1) = (3, 1) \cdot 2D; \\ (Z)\{3(4) - (2, 2) - (3, 1)\} \\ &+ \{2(2, 2) - (2, 2)(2m - 5) - (1, 1, 2)\} + \text{Supp. } (2, 2) = (2, 2) \cdot 2D; \\ (Z)\{2(3, 1) - (2, 1, 1) \cdot 2 - (2, 1, 1) \cdot 2\} \\ &+ \{(2, 1, 1) - (2, 1, 1)(2m - 6) - (1, 1, 1, 1) \cdot 3\} \\ &+ \text{Supp. } (2, 1, 1) = (2, 1, 1) \cdot 2D; \\ (Z)\{4(4) - (1, 3) - (4)(2m - 4)\} \\ &+ \{(1, 3) - (1, 3)(2m - 5) - (1, 3)(2m - 5)\} + \text{Supp. } (1, 3) = (1, 3) \cdot 2D; \end{aligned}$$

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$$\begin{aligned} (Z)\{3(3, 1) - (1, 1, 2) - (3, 1)(2m - 5)\} \\ &+ 2\{2(2, 2) - (1, 1, 2) - (2, 2)(2m - 5)\} \\ &+ \{2(1, 1, 2) - (1, 1, 2)(2m - 6) - (1, 1, 2)(2m - 6)\} \\ &+ \text{Supp. } (1, 1, 2) = (1, 1, 2) \cdot 2D; \\ (Z)\{2(2, 1, 1) - (1, 1, 1, 1) \cdot 3 - (2, 1, 1)(2m - 6)\} \\ &+ \{4(1, 1, 1, 1) - (1, 1, 1, 1)(2m - 7) - (1, 1, 1, 1)(2m - 7)\} \\ &+ \text{Supp. } (1, 1, 1, 1) = (1, 1, 1, 1) \cdot 2D. \end{aligned}$$

110. I content myself with giving the expressions of only the following supplements.

$$\begin{aligned}\text{Supp. } (4Z)(1) &= m[2(\cdot) - (\cdot/)] + \kappa(\cdot), \\ \text{Supp. } (3Z)(2) &= \frac{1}{2}m[2(\cdot) - (\cdot/)] + \frac{1}{2}\kappa(\cdot/), \\ \text{Supp. } (3Z)(1, 1) &= (2mn - 3n^2 - m + n + \alpha)(\cdot) \\ &\quad + (2m^2 - 4mn - 2m + 2n + (m - n)\alpha)(\cdot/) \\ &\quad + (-m^2 + m\alpha)(//), \\ \text{Supp. } (2Z)(3) &= -\frac{1}{2}m[2(\cdot) - (\cdot/)] \\ &\quad + \frac{1}{2}n[2(\cdot) - (\cdot/) + 2(2(\cdot/) - (//))] + \frac{1}{2}\kappa(\cdot/).\end{aligned}$$

where $(4Z)$ stands for any one of the symbols (4) , $(3, 1)$, $\dots$, $(1, 1, 1, 1)$.

111. The expression of Supp. $(4Z)(1)$ has been explained supra, No. 108. That of Supp. $(3Z)(2)$ may also be explained.

1°. The point-pairs of the system of conics $(3Z)$, regarding each point-pair as a line, are a set of lines enveloping a curve; the class of this curve is equal to the number of the lines which pass through an arbitrary point, that is, as at first sight would appear, to the number of point-pairs in the system $(3Z\cdot)$, or to $2(3Z) - (3Z/)$: it is, however, necessary to admit that the number of distinct lines, and therefore the class of the curve, is one-half of this, or $= \frac{1}{2}[2(3Z) - (3Z/)]$; which being so, the number of the point-pairs $(3Z)$ which, regarded as lines, touch the given curve (of the order m and class n) is $= \frac{1}{2}n[2(3Z) - (3Z/)]$.

The point of contact of any one of these lines with the given curve is (specially) a united point, and we have thus the term $\frac{1}{2}n[2(3Z) - (3Z/)]$ of the Supplement.

2°. The number of the conics $(3Z)$ which touch the given curve at a given cusp thereof, or, say, the conics $(3Z)(2\kappa l)$, is $= \frac{1}{2}(3Z/)$, and the cusp is in respect of each of these conics a united point; we have thus the remaining term $\frac{1}{2}\kappa(3Z/)$ of the Supplement.

Article Nos. 112 to 135. Application to the Conies which satisfy five conditions of contact with a given Curve.

112. We have twelve equations, which I first present in what I call their original forms; viz. these are
First equation:

$$\{(5) - (5)(2m - 5) - (1, 4)\} + \text{Supp. } (5) = 5(5) \cdot 2D.$$

Second equation:

$$2\{(5) - (4, 1) - (2, 3)\} + \{(4, 1) - (4, 1)(2m - 6) - (1, 1, 3)\} + \text{Supp. } (4, 1) = 4(4, 1) \cdot 2D.$$

Third equation:

$$3\{(5) - 2(3, 2)\} + \{(3, 2) - (3, 2)(2m - 6) - 2(1, 2, 2)\} + \text{Supp. } (3, 2) = 3(3, 2) \cdot 2D.$$

Fourth equation:

$$2\{(4, 1) - 2(3, 1, 1) - (2, 2, 1)\} + \{(3, 1, 1) - (3, 1, 1)(2m - 7) - (1, 1, 1, 2)\} + \text{Supp. } (3, 1, 1) = 2(3, 1, 1) \cdot 2D.$$

Fifth equation:

$$4\{(5) - (2, 3) - (4, 1)\} + \{(3, 2) - (2, 3)(2m - 6) - (1, 1, 3)\} + \text{Supp. } (2, 3) = 2(2, 3) \cdot 2D.$$

Sixth equation:

$$3\{(4, 1) - 2(2, 2, 1) - (3, 1, 1)\} + 2\{(3, 2) - (2, 2, 1) - (2, 2, 1)\} + \{2(2, 2, 1) - (2, 2, 1)(2m - 7) - 2(1, 1, 1, 2)\} + \text{Supp. } (2, 2, 1)$$

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Seventh equation:

$$2\{(3, 1, 1) - 3(2, 1, 1, 1)\} + \{(2, 1, 1, 1) - (2, 1, 1, 1)(2m-8) - 4(1, 1, 1, 1, 1)\} + \text{Supp. } (2, 1, 1, 1) = 2(2, 1, 1, 1) \cdot 2D.$$

Eighth equation:

$$5\{(5) - (1, 4) - (5)(2m-5)\} + \{(4, 1) - (1, 4)(2m-5) - (1, 4)(2m-5)\} + \text{Supp. } (1, 4) = (1, 4) \cdot 2D.$$

Ninth equation:

$$4\{(4, 1) - (1, 1, 3) - (4, 1)(2m-6)\} + 2\{(3, 2) - (1, 1, 3) - (2, 3)(2m-6)\} + \{2(3, 1, 1) - (1, 1, 3)(2m-7) - (1, 1, 3)(2m-7)\} +$$

Tenth equation:

$$3\{(3, 2) - 2(1, 2, 2) - (3, 2)(2m-6)\} + \{(2, 2, 1) - (1, 2, 2)(2m-7) - (1, 2, 2)(2m-7)\} + \text{Supp. } (1, 2, 2) = (1, 2, 2) \cdot 2D.$$

Eleventh equation:

$$3\{(3, 1, 1) - (1, 1, 1, 2) - (3, 1, 1)(2m-7)\} + 2\{2(2, 2, 1) - 2(1, 1, 1, 2) - (2, 2, 1)(2m-7)\} + \{3(2, 1, 1, 1) - (1, 1, 1, 2)(2m-8)\}$$

Twelfth equation:

$$2\{(2, 1, 1, 1) - 4(1, 1, 1, 1, 1) - (2, 1, 1, 1)(2m-8)\} + \{5(1, 1, 1, 1, 1) - (1, 1, 1, 1, 1)(2m-9) - (1, 1, 1, 1, 1)(2m-9)\} + \text{Supp.}$$

113. I alter the forms of these equations by substituting for $2D$ its value $= n - 2m + 2 + \kappa$, and by writing for the expressions with $(\bar{1})$ their values, $(\bar{1}, 4) = (\cdot 4) - 6(5)$, &c., and except in the terms $\{\text{Supp. } (5) - \kappa(5)\}$, &c., by writing for κ its value $= -3m + \alpha$. The resulting equations, if the Supplements were known, would serve to determine the values of (5) , $(4, 1)$, &c.; but I assume instead that the last-mentioned expressions are known (First Memoir, No. 50), and use the equations to determine the Supplements, or, what comes to the same thing, the values of the terms in $\{ \}$ which contain these Supplements.

We have thus the twelve reduced equations, with resulting values of the supplements.

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114. First equation:

$$\begin{aligned} &= (5) - 15m - 15n + 9\alpha \\ &+ \{\text{Supp. } (5) - \kappa(5)\} - 3m + \alpha \\ &+ (5)(8m + 7n - 4\alpha) \\ &- (\bar{1}, 4) + 10m + 8n - 6\alpha. \end{aligned}$$

(that is, we have $\text{Supp. } (5) - \kappa(5) = -3m + \alpha$, and so in the subsequent cases, the equation gives the value of the term in $\{ \}$ which contains the Supplement).

115. Second equation:

$$\begin{aligned} &= 2(5) - 30m - 30n + \alpha(18) \\ &+ (4, 1) - 8m^2 - 20mn - 8n^2 + 104m + 104n + \alpha(6m + 6n - 66) \\ &+ \{\text{Supp. } (4, 1) - \kappa(4, 1)\} - 6m^2 - 3mn + 18m + 9n + \alpha(3m - 9) \\ &+ (4, 1)(6m + 5n - 3\alpha) \\ &- (\bar{1}, 3) + 8m^2 + 12mn + 8n^2 - 56m - 53n + \alpha(-6m - 3n + 39). \end{aligned}$$

I stop for a moment to notice a very convenient verification of the term in $\{ \}$; putting therein $\alpha = 3n$, the term is $-6m^2 - 3mn + 18m + 9n + (9mn - 27n)$; and if in this we write $m^2 = mn = n^2 = 2$,

and when any higher terms enter $m^3 = m^2n = mn^2 = n^3 = 4$, $m^4 = m^3n = m^2n^2 = mn^3 = n^4 = 8$, &c., the value is $-12 - 6 + 18 + 9 + 18 - 27 = 0$, viz. we should always obtain a sum = 0. The reason is that the term in question should always admit of being

expressed in the form $p\delta + q\kappa + r\tau + s\iota$; the reduction to this form might be effected by the substitutions $m = \frac{1}{2}(m+n) + \frac{1}{2}(\kappa - \iota)$, $n = \frac{1}{2}(m+n) - \frac{1}{2}(\kappa - \iota)$, $m^2 = 2 \cdot \frac{1}{2}(m+n) + 2\delta + 3\kappa$, $n^2 = 2 \cdot \frac{1}{2}(m+n) + 2\tau + 3\iota$, giving a result = $A(m+n) + \text{terms in } (\delta, \kappa, \tau, \iota)$, where A is a numerical coefficient calculable as above by simply writing $m = n = 1$, $m^2 = mn = n^2 = 2$, &c., and which is = 0 when the term is of the proper form $p\delta + q\kappa + r\tau + s\iota$. The complete reduction to the form in question is material in the sequel, but I advert to the point here only for the sake of the numerical verification.

116. Third equation:

$$\begin{aligned} &= 3(5) - 60m - 60n + \alpha(36) \\ &+ (3, 2) - 8m^2 - 20mn - 8n^2 + 120m + 120n + \alpha(-4m - 4n - 78) + 3\alpha^2 \\ &+ \{\text{Supp. } (3, 2) - \kappa(3, 2)\} - 15m^2 - 36mn - 27n^2 + \alpha(4m + 3n + 18) - 2\alpha^2 \\ &+ (3, 2)(4m + 3n - 2\alpha) \\ &- 2(\bar{2}, 2) - 54m^2 - 48n^2 + \alpha(+40) - \alpha^2. \end{aligned}$$

Verification is $16 + 3(1 \cdot 2 - 7) = 0$.

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117. Fourth equation:

$$= 2(4, 1) - 16m^2 - 40mn - 16n^2 \tag{1}$$

$$+ (3, 1, 1) - \frac{1}{2}m^3 - 10m^2n - 10mn^2 - \frac{1}{2}n^3 + 59m^2 + 116mn + 59n^2 \tag{2}$$

$$+ \{\text{Supp. } (3, 1, 1) - \kappa(3, 1, 1)\} - \frac{3}{2}m^3 + \frac{3}{2}m^2n + \frac{3}{2}mn^2 + \frac{3}{2}n^3 - 26m^2 - \frac{51}{2}mn - \frac{25}{2}n^2 \tag{3}$$

$$+ (3, 1, 1)(4m + 3n - 2\alpha) \tag{4}$$

$$- (\bar{2}, 1, 1) - 24m^2 - 36mn - 12n^2 \tag{5}$$

$$(1) \quad + 208m + 208n + \alpha(+12m + 12n - 132)$$

$$(2) \quad - 434m - 434n + \alpha(\frac{1}{2}m^2 + 6mn + \frac{1}{2}n^2 - 62m - 62n + 291) - \frac{3}{2}\alpha^2$$

$$(3) \quad - 507m - 253n + \alpha(\frac{3}{2}m^2 - \frac{3}{2}m - \frac{3}{2}n + 33)$$

$$(4) \quad + 108m + 81n + \alpha(-m^2 - 4mn - n^2 + 7m + \frac{7}{2}n - 54) + 3\alpha^2$$

$$(5) \quad + 168m + 168n + \alpha(-m^2 - 2mn - \frac{1}{2}n^2 + 25m + \frac{25}{2}n - 138) + \frac{1}{2}\alpha^2.$$

Page 276**119.** Sixth equation:

$$= 3(4, 1) + 2(3, 2) + 2(2, 2, 1) \quad (1,2,3)$$

$$+ \{\text{Supp. } (2, 2, 1) - \kappa(2, 2, 1)\} \quad (4)$$

$$+ (2, 2, 1)(2m + n - \alpha) - 2(\bar{2}, 1, 1) \quad (5,6)$$

$$= -24m^2 - 60mn - 24n^2 + 312m + 312n \quad (1)$$

$$+ 240m + 240n \quad (2)$$

$$+ 48m^2 + 108mn + 48n^2 - 936m - 936n \quad (3)$$

$$+ 12m^2 + 6mn - 6n^2 - 60m - 6n \quad (4)$$

$$+ 12m^2 + 18mn + 6n^2 + 108m + 54n \quad (5)$$

$$- 48m^2 - 72mn - 24n^2 + 336m + 336n \quad (6)$$

$$(1) \quad + \alpha(+18m + 18n - 198)$$

$$(2) \quad + \alpha(-8m - 8n - 156) + \alpha^2(+6)$$

$$(3) \quad + \alpha(-16m - 16n + 654) + \alpha^2(m + n - 24)$$

$$(4) \quad + \alpha(mn - 8m - 2n + 30)$$

$$(5) \quad + \alpha(2m^2 + 3mn + n^2 - 36m - 21n - 6) + \alpha^2(-m - n + 15)$$

$$(6) \quad + \alpha(-2m^2 - 4mn - n^2 + 50m + 29n - 276) + \alpha^2(+3).$$

Verification is $(12 + 6 - 6)2 - 60 - 6 + 3(4 - (8 + 2)2 + 30) = 0$.**Page 277****121.** Eighth equation:

$$= 5(5) + (4, 1)$$

$$+ \{\text{Supp. } (\bar{1}, 4) - \kappa(\bar{1}, 4)\}$$

$$- (m - \frac{1}{2})\{2(\cdot 4) - (/4)\}$$

$$= -75m - 75n + \alpha(45)$$

$$- 8m^2 - 20mn - 8n^2 + 104m + 104n + \alpha(6m + 6n - 66)$$

$$+ 3m - 4n + \alpha(1)$$

$$+ (-n + \frac{1}{2})(\cdot 4) + (-m + \frac{1}{2})(/4) + \alpha n(5).$$

Verification is $1 - 4 + 3 \cdot 1 = 0$.**122.** Ninth equation:

$$= 4(4, 1) + 2(3, 2) + 2(3, 1, 1)$$

$$+ \{\text{Supp. } (\bar{1}, 1, 3) - \kappa(\bar{1}, 1, 3)\}$$

$$- (m - 2)\{2(\cdot 1, 3) - (/1, 3)\}$$

$$+ (-n + 2)(\cdot 1, 3) + (-m + 2)(/1, 3) + n(4(4, 1) + 2(2, 3))$$

Page 278**123.** Tenth equation:

$$= 3(3, 2) + (2, 2, 1)$$

$$+ \{\text{Supp. } (\bar{1}, 2, 2) - \kappa(\bar{1}, 2, 2)\}$$

$$- (m - 2)\{2(\cdot 2, 2) - (/2, 2)\}$$

$$+ (-n + 2)(\cdot 2, 2) + (-m + 2)(/2, 2) + 3n(3, 2)$$

124. Eleventh equation:

$$\begin{aligned}
 &= 3(3, 1, 1) + 4(2, 2, 1) + 3(2, 1, 1, 1) \\
 &+ \{\text{Supp. } (\bar{1}, 1, 1, 2) - \kappa(\bar{1}, 1, 1, 2)\} \\
 &\quad - (m - \frac{5}{2})\{2(\cdot 1, 1, 2) - (/ 1, 1, 2)\} \\
 &\quad + (-n + \frac{5}{2})(\cdot 1, 1, 2) + (-m + \frac{5}{2})(/ 1, 1, 2) \\
 &\quad + n(3(3, 1, 1) + 2(2, 2, 1))
 \end{aligned}$$

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125. Twelfth equation:

$$\begin{aligned}
 &= 2(2, 1, 1, 1) + 5(1, 1, 1, 1, 1) \\
 &+ \{\text{Supp. } (\bar{1}, 1, 1, 1, 1) - \kappa(\bar{1}, 1, 1, 1, 1)\} \\
 &\quad - (m - \frac{5}{2})\{2(\cdot 1, 1, 1, 1) - (/ 1, 1, 1, 1)\} \\
 &\quad + (-n + \frac{5}{2})(\cdot 1, 1, 1, 1) + (-m + \frac{5}{2})(/ 1, 1, 1, 1) \\
 &\quad + 2n(2, 1, 1, 1).
 \end{aligned}$$

The complete expressions for the supplements obtained from these twelve equations are tabulated in the following articles.

126–132. [The detailed expressions for the twelve supplements are given in the following articles, with the resulting values:]

$$\begin{aligned}
 \text{Supp. (5)} &= N + O, \\
 \text{Supp. (4, 1)} &= m[2(\cdot) - (/)] + \kappa(\cdot) + \dots, \\
 \text{Supp. (3, 2)} &= \dots, \\
 &\&c.,
 \end{aligned}$$

where $N (= \iota)$ is the number of line-pair-points described as “inflexion tangent terminated each way at inflexion,” and $O (= \kappa)$ the number of line-pair-points described as “cuspidal tangent terminated each way at cusp.”

127.

$$\begin{aligned}
 A &= \frac{1}{2}\iota(m-4)(m-5) = \frac{1}{2}m^2 + 2m^2 - \frac{1}{2}m^3 - \frac{1}{2}m^2n^2 - 18m^2 + \frac{1}{2}mn + 5n^2 + 40m - 5n + \alpha(-\frac{1}{2}m^2 + \frac{1}{2}m - 10); \\
 B &= \frac{1}{2}\iota(m-3)(m-4) = -\frac{1}{2}m^3 + \frac{1}{2}m^2 - 18m + \alpha(\frac{1}{2}m^2 - \frac{1}{2}m + 6); \\
 C &= \frac{1}{2}\iota(m-4)(m-5) = \frac{1}{2}N + 2m^2 - 4m^2 - \frac{1}{2}m^2n^2 - 18m^2 + \frac{1}{2}mn + 5n^2 + 40m - 5n + \alpha(-\frac{1}{2}m^2 + \frac{1}{2}m - 10); \\
 D &= \frac{1}{2}\iota(m-3)(m-4) = -\frac{1}{2}m^3 + \frac{1}{2}m^2 - 18m + \alpha(\frac{1}{2}m^2 - \frac{1}{2}m + 6).
 \end{aligned}$$

128. I make the following calculations, serving to express in terms of Zeuthen’s Capitals, the terms in { } contained in the twelve equations respectively.

$$\begin{aligned}
 N &= -3m + \alpha \quad (\text{first equation}). \\
 2J &= -6m^2 + 18m + 9n + \alpha(2m - 6), \\
 R &= -3mn + \alpha(m - 3), \\
 6K &= \dots \\
 L &= \dots \\
 &= -6m^2 - 3mn + 18m + 9n + \alpha(3m - 9) \quad (\text{second equation}). \\
 &\vdots \\
 &= -\frac{1}{2}m^3 + \frac{3}{2}m^2n + 2mn^2 + \frac{1}{2}n^3 + \frac{9}{2}m^2 - 7n^2 - 50m - 23n + \alpha(\frac{1}{2}m^2 - \frac{1}{2}m - \frac{1}{2}n + 33) \quad (\text{fourth equation}).
 \end{aligned}$$

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$$\begin{aligned} Q &= n^2 + 8m - n - 3\alpha \quad (\text{fifth equation}). \\ 3G &= \dots \\ &= 12m^2 + 6mn - 6n^2 - 60m - 6n + \alpha(mn - 3m - 2n + 30) \quad (\text{sixth equation}). \end{aligned}$$

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129. We have consequently, by means of the results just obtained,

$$\begin{aligned} \text{Supp. (5)} &= \kappa(5) + N, \\ \text{Supp. (4, 1)} &= \kappa(4, 1) + 2J + R, \\ \text{Supp. (3, 2)} &= \kappa(3, 2) + 6K + L + 3N + 2O, \\ \text{Supp. (3, 1, 1)} &= \kappa(3, 1, 1) + D + E + F + 2G + 2J + J', \\ \text{Supp. (2, 3)} &= \kappa(2, 3) + Q, \\ \text{Supp. (2, 2, 1)} &= \kappa(2, 2, 1) + 3G + I + 4J + 2J', \\ \text{Supp. (2, 1, 1, 1)} &= \kappa(2, 1, 1, 1) + B + 4D + 4D' \quad (\text{seventh equation}). \end{aligned}$$

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$$\begin{aligned} \text{Supp. } (\bar{1}, 4) &= \kappa(\bar{1}, 4) + (n - \tfrac{1}{2})(4N + 2O) - \tfrac{1}{2}N + \tfrac{3}{2}O \quad (\text{eighth equation}), \\ \text{Supp. } (\bar{1}, 1, 3) &= \dots \quad (\text{ninth equation}), \\ \text{Supp. } (\bar{1}, 2, 2) &= \dots \quad (\text{tenth equation}), \\ \text{Supp. } (\bar{1}, 1, 1, 2) &= \dots \quad (\text{eleventh equation}), \\ \text{Supp. } (\bar{1}, 1, 1, 1, 1) &= \dots \quad (\text{twelfth equation}). \end{aligned}$$

130–132. [The remaining supplements are calculated in a similar manner, and the complete table of results is given.]

133. These are, I think, the true theoretical forms of the Supplements, viz. (attending to the signification of the Capitals) the expressions actually exhibit how the Supplement arises, whether from proper conics passing through or touching at a cusp, or from point-pairs (coincident line-pairs) or line-pairs (including of course in these terms line-pair-points). Thus, for instance,

$$\text{Supp. (5)} = N + O.$$

Referring to the explanations, First Memoir, Nos. 41 to 47, $N(= \iota)$ is the number of the line-pair-points described as “inflexion tangent terminated each way at inflexion,” and $O(= \kappa)$ the number of the line-pair-points described as “cuspidal tangent terminated each way at cusp,” or in what is here the appropriate point of view, we have as a coincident line-pair each inflexion tangent and each cuspidal tangent. Reverting to the generation of the first equation, when the point P is a point in general of the given curve, the curve Θ is the conic (5), having with the curve 5 intersections at P , and besides meeting it in the $2m - 5$ points P' . When the point P is at an inflexion, the curve Θ becomes the coincident line-pair formed by the tangent taken twice, the number of intersections at P is therefore $= 6$, and the inflexion is therefore (specially) a united point. Similarly, when the point P is at a cusp, the curve Θ becomes the coincident line-pair formed by the tangent taken twice, the number of intersections at P is therefore $= 6$, and the cusp is thus (specially) a united point: we have thus the total number of special united points $= \kappa + \iota$, agreeing with the foregoing *a posteriori* result, $\text{Supp. (5)} = N + O$.

134. Or to take another example; for the fifth equation we have

$$\text{Supp. } (\bar{2}, 3) = \kappa(\bar{2}\kappa\bar{l}, 3) + Q;$$

Q ($= 2\tau$) is the number of the line-pair-points described as “double tangent terminated each way at point of contact,” or, in the point of view appropriate for the present purpose, we have each double tangent as a coincident line-pair in respect to the one of its points of contact, and also as a coincident line-pair in respect to the other of its points of contact. Reverting to the generation of the equation, when the point P is a point in general on the given curve, the curve Θ is the system of conics $(\bar{2}, 3)$ touching the curve at P , and having besides with it a contact of the third order; since for each conic the number of intersections at P is $= 2$, the total number of intersections at P is $= 2(\bar{2}, 3)$, and the remaining $(2m - 2)(\bar{2}, 3)$ intersections are the points P' . Suppose that the point P is taken at the point of contact of a double tangent; of the $(\bar{2}, 3)$ conics, 1 (I assume this is so) becomes the coincident line-pair formed by the double tangent taken twice, and gives therefore 4 intersections at P , the remaining $(\bar{2}, 3) - 1$ conics are proper conics, giving therefore $2(\bar{2}, 3) - 2$ intersections at P , or the total number of intersections at P is $2(\bar{2}, 3) + 2$ intersections; or there is a gain of 2 intersections. As remarked (No. 96), this does not of necessity imply that the point in question is to be considered as being (specially) 2 united points; I do not know how to decide *a priori* whether it is to be regarded as being 2 united points or as 1 united point, but it is in fact to be regarded as being (specially) only 1 united point; and as the points in question are the 2τ points of contact of the double tangents, we have thus the number 2τ of special united points. Again, when the point P is at a cusp, all the $(\bar{2}, 3)$ conics remain proper conics $((2\kappa l, 3) = (\bar{2}, 3))$,

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and we thus obtain the foregoing *a posteriori* result $\text{Supp. } (\bar{2}, 3) = Q$.

135. [Further discussion of the theoretical forms of the remaining supplements.]

Paper [408]

Addition to Memoir on the Resultant of a System of Two Equations.

[PAGES 296–299]

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408. [From the Philosophical Transactions of the Royal Society of London, vol. CLVIII. (for the year 1868), pp. 579–588.]

This is an addition to the memoir on the resultant of a system of two equations, giving further developments of the theory established in the paper [380]. The memoir contains tables for the calculation of resultants for systems of equations of various degrees.

Table (4, 4) continued:

[The page contains tabular values for the resultant of two quartic equations, with numerical coefficients arranged in a systematic scheme for computing the resultant of two quartics.]

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Table (4, 4) concluded:

[These pages contain the conclusion of the table for the resultant of two quartic equations, completing the systematic enumeration of the numerical coefficients. The volume of tabular material is too extensive to reproduce here in full.]

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[The page concludes the addition to the memoir on resultants, completing the discussion of the tabular values for systems of two equations.]

Paper [409]

On the Conditions for the Existence of Three Equal Roots,
or of Two Pairs of Equal Roots, of a Binary Quartic or Quintic.

[PAGE 300]

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLVIII. (for the year 1868), pp. 577–588. Received November 26, 1867,—Read January 9, 1868.]

409.

It is remarked, Proc. R. Soc. vol. xvii. p. 314, that the above title is a misnomer: I had in fact in regard to the quintic considered not the twofold relations belonging to the root-systems 311 and 221 respectively, but the threefold relations belonging to the root-systems 41 and 32 respectively. The proper title would have been “On the conditions for the existence of certain systems of equal roots of a binary quartic or quintic.”

In considering the conditions for the existence of given systems of equalities between the roots of an equation, we obtain some very interesting examples of the composition of relations. A relation is either onefold, expressed by a single equation $U = 0$, or it is, say k -fold, expressed by a system of k or more equations. Of course, as regards onefold relations, the theory of the composition is well known: the relation $UV = 0$ is a relation compounded of the relations $U = 0$, $V = 0$; that is, it is a relation which is satisfied if either of the component relations is satisfied. And so in the case of a twofold relation, expressed by the two equations $U = 0$, $V = 0$, if these equations contain each of them a factor, say if $U = PW$, $V = QW$, then the twofold relation $U = 0$, $V = 0$ is equivalent to the onefold relation $W = 0$, together with the twofold relation $P = 0$, $Q = 0$; the twofold relation is thus compounded of the onefold relation and a twofold relation. I do not in the present memoir attempt any general theory of the composition of relations; but I establish in a direct manner certain identities which put in evidence the composition of the relations in the cases of a binary quartic or quintic which present themselves in the theory.

The Binary Quartic.

Article Nos. 1 to 11.

1. Let the equation be $(a, b, c, d, e \chi x, y)^4 = 0$, then we have

$$\begin{aligned} I &= ae - 4bd + 3c^2, \\ J &= ace + 2bcd - ad^2 - b^2e - c^3, \end{aligned}$$

and the discriminant is $\Delta = I^3 - 27J^2$. When $\Delta = 0$, the equation has two equal roots. For the existence of three equal roots we have the twofold relation expressed by the two equations

$$I = 0, \quad J = 0.$$

It is to be observed that the condition $I = 0$ implies that the quartic can be expressed as the sum of two fourth powers, and the condition $J = 0$ is then the condition that these two fourth powers have a common root.

2. The further development continues with the detailed conditions for the existence of two pairs of equal roots, and the corresponding investigation for the binary quintic; establishing by direct calculation

certain identities which put in evidence the composition of the relations in the several cases that present themselves in the theory.